Akshat G

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Summary

Computer Science student, AI engineer, and MedTech startup co-founder with experience building production ML systems, scalable backend infrastructure, and mobile applications. Skilled in GPU-accelerated inference, computer vision, LLM deployment, and distributed AI pipelines from research to production. Experience leading technical teams and delivering end-to-end software systems used by real users.

Education

Manipal Institute of Technology — Bengaluru, India
Bachelor of Technology (Honours) — Computer Science and Engineering — *Jul 2023 – May 2027*

- **Specialization:** Artificial Intelligence
- **CGPA:** 9.56/10

Experience

Shell India Markets Private Limited — Bengaluru, India
Intern, Multiphysics AI and Product Innovation — *Jun 2026 – Present*

- Designed **HiME**, a hierarchical multilabel classifier for gear failure analysis, achieving **100% hierarchy consistency** and **90% expert accuracy** by enforcing hierarchical constraints during training rather than post-processing.
- Developed a modular inference framework that decoupled feature extraction from hierarchical classification, accelerating experimentation with multiple vision backbones.
- Deployed an automated gear failure assessment system that reduced manual report generation, allowing domain experts to focus on high-value human-in-the-loop annotation and validation.

Sensera — Bengaluru, India
Co-Founder & ML Engineer — *Feb 2026 – Jun 2026*

- Co-founded an AI MedTech startup building intelligent Personal Health Records; built and shipped a full-stack platform supporting medication tracking, vitals, and document management for **50+ daily active users**.
- Architected an asynchronous document-processing pipeline that decoupled OCR and NLP inference across **10 parallel GPU workers**, reducing end-to-end latency to **3 s warm-start** and **10 s cold-start**.
- Implemented semantic search over medical records using pgvector and row-level security, enabling sub-second retrieval while isolating user data.
- Deployed a self-hosted LLM pipeline for summarization and structured information extraction, eliminating reliance on external inference providers while keeping sensitive patient data within self-hosted infrastructure.
- Core ML infrastructure and proprietary medical AI assets acquired following product development.

CEAM - Center of Excellence in Autonomous Mobility — Bengaluru, India
Machine Learning Lead – Code — *Aug 2025 – Aug 2026*

- Led a **16-member** computer vision team developing Project MAVERIC, an autonomous perception stack for real-time robotic navigation.
- Engineered high-performance perception pipelines using optimized C++ (Eigen) and NumPy, delivering real-time semantic segmentation (**56 Hz**) and object detection (**16 FPS**).
- Developed a custom LiDAR-camera fusion pipeline enabling accurate 3D scene understanding and supporting real-time path planning.

Technical Skills

- **Programming:** Python, TypeScript/JavaScript, Java, C, SQL, Bash
- **ML Frameworks & Tools:** PyTorch, TensorFlow, HuggingFace Transformers, Sentence-Transformers, TensorRT, ONNX Runtime, OpenCV, scikit-learn
- **Core Expertise:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Generative AI
- **AI/ML Specialization:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Dense Retrieval, Semantic Search, Vector Embeddings, Transfer Learning, Fine-tuning
- **Distributed Systems & Cloud:** Docker, Modal (Serverless GPU), Supabase, Google Cloud (OAuth, Drive API), AWS
- **Databases:** PostgreSQL (Schema Design, Row-Level Security, Migrations, Indexing), pgvector (HNSW ANN)
- **Frontend & Backend:** React Native (Expo SDK 54), React, FastAPI, Pydantic

Publications

Authored **8+** publications in data-centric AI, semi-supervised learning, and few-shot learning. Published in *Scientific Reports*, *SAGE Digital Health*, and Springer CCIS, peer reviewed in *IEEE Transactions on Medical Imaging*. Google Scholar